QUOTIENTS BY A FINITE GROUP ACTION INHERIT QUASI-SEPARATEDNESS AND SEPARATEDNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement inherits the notation of the preceding quotient lemma through a reference; the reference is recorded as a non-load-bearing navigation pointer while the counted statement stays self-contained in wording.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The quotient by a group action is an etale equivalence relation. Lemma lemma-quotient (2463-2487, produced as L44177), used at proof lines 2549-2550. Supplies the notation and assumptions.
- Reduction of diagonal properties of a quotient to the map j. Lemma lemma-properties-diagonal (2296-2308, produced as L44181), used at proof lines 2559-2561. Reduces the claims to properties of j.

Source reference. Original lemma 02X4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3827–3827. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02Z2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3897–3897. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Notation and assumptions as in Lemma 02Z2. Assume G is finite. Then*

(1) *if $U \to S$ is quasi-separated, then U/G is quasi-separated over S, and*
(2) *if $U \to S$ is separated, then U/G is separated over S.*

Proof. In the proof of Lemma 02X4 we saw that it suffices to prove the corresponding properties for the morphism $j : R \to U \times_S U$. If $U \to S$ is quasi-separated, then for every affine open $V \subset U$ which maps into an affine of S the opens $g(V) \cap V$ are quasi-compact. It follows that j is quasi-compact. If $U \to S$ is separated, the diagonal $\Delta_{U/S}$ is a closed immersion. Hence $j : R \to U \times_S U$ is a finite coproduct of closed immersions with disjoint images. Hence j is a closed immersion. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.